



STUDY PLAN

Master of Applied IT

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HBO-i Architectural Layers

Level 4 Domain: Organisational Processes

Level 3 Domain: Software Engineering

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Background

I am an analytical and results-driven IT professional with a passion for leveraging data and technology to drive organisational success. With hands-on experience in data analysis, process optimisation, and dashboard development across companies like ASML, Ambianti, and Hunkemöller, I thrive at the intersection of technology and business operations. I am proficient in tools such as Power BI, SQL, and Python, and certified in Lean Six Sigma Green Belt. I focus on delivering actionable insights and sustainable process improvements. Known for my proactive approach, leadership in team environments, and effective communication skills, I am dedicated to continuously growing my expertise in organisational processes and software design to make a meaningful impact in every project I undertake.

Background within the IT domain

- **Organisational Processes:** Through roles at ASML, Ambianti, and my current position at Hunkemöller, I have systematically analysed organisational processes, identified inefficiencies, and led optimisation initiatives. For instance, at ASML, I automated manual tasks, which improved KPI tracking by 20%. At Ambianti, I led process optimisation initiatives, resulting in measurable efficiency improvements. Currently, I support data governance at Hunkemöller to ensure reliable data for process decisions.
- **Software:** I have built various software solutions and dashboards using C#, Java for full-stack development; Python, SQL, Looker for data processing and visualization (ASML, Ambianti) and GCP, Terraform and BigQuery for data engineering (Hunkemöller).

Thus, my self-assessment before the start of semester 1 would be:

Domain	ANALYSIS	ADVISE	DESIGN	REALISE	MANAGE & CONTROL
<i>Organisational Processes</i>	Level 3	Level 3	Level 3	Level 3	Level 3
<i>Software</i>	Level 2	Level 2	Level 2	Level 2	Level 2

Learning Goals for the Master of Applied IT

In the master program, I want to sharpen my ability to connect business goals to technology outcomes. Thus, I want to focus on two domains: Organisational Processes (Level 4), where I will shape end-to-end data architecture, and Software (Level 3), where I will deepen development practices for AI-enabled systems. Building on my experience, I will analyse current and target states, advise stakeholders, and design pragmatic roadmaps. Thus, I would like to make the following growth related to the HBO-I architectural pillars:

- **Organisational Processes – Level 4:** I want to focus on building end-to-end data architectures aim to align technology decisions with business outcomes. Drawing on my experience in process optimisation and data products at ASML, Ambianti, and Hunkemöller, I will focus on analysing current and target states, advising stakeholders in clear terms, and designing architectures that enable reliable data flows, secure integrations, and scalable delivery. I aim to explore options objectively and document trade-offs, backed by theoretical academic studies, using lightweight governance (standards, ADRs) to keep decisions transparent. Leveraging my skills in SQL, Python and my exposure to GCP, Terraform, and BigQuery, I want to craft architectures that support trustworthy analytics and allow for incremental changes. My goal is to design proofs of concepts and roadmaps that show early value, so the resulting architecture will aim to be purposeful, explainable, and executable across the organisation.
- **Software - Level 3:** I would like to deepen my skills in software development and software architecture for solutions, focusing on analysis, advising stakeholders, and designing robust applications. Especially with the current rise in popularity of AI and Machine Learning algorithms, I would like to dedicate time in learning and improving in the software layer especially when it comes to development of AI and AI powered tools and pipelines.

Thus, my development goal for the end of semester 1 would be:

Domain	ANALYSIS	ADVISE	DESIGN	REALISE	MANAGE & CONTROL
<i>Organisational Processes</i>	Level 4	Level 4	Level 4	Level 4	Level 4
<i>Software</i>	Level 3	Level 3	Level 3	Level 3	Level 3

Research

My will explore the ways in which semi-supervised learning combined with synthetic dermatological images can make skin-cancer detection AI more cost-efficient and potentially remove known biases from the currently available classification models. The project is embedded in the Fontys Research Group “Sustainable Data & AI Applications”, with Qin Zhao as project owner/supervisor, and Ralf Raumanns as domain expert for dermatology-AI validation.

The main research question is: *“How can semi-supervised learning and synthetic data help achieve cost-efficient, sustainable AI software solutions for skin-cancer detection?”*

I aim to answer it by making a prototype of an end-to-end pipeline, from synthetic data generation and weak/SSL training to evaluation on real-only test sets, while documenting decision rationales. The aim is that the work will empathize on already available measurable acceptance criteria, form the available studies of Ralf Ra Raumanns, together with fairness checks, and bias mitigation from the aim to generate more balanced synthetic data focusing on a wider range of skin tones, thus with this reflecting well-known representation challenge with the available dermatology datasets, containing more pale skin tones, rather than having a more balanced distribution with darker ones.

Learning goals for the first semester

As mentioned in the sections above I will aim to develop Level 4 competency in Organisational Processes and Level 3 competence in Software. At Level 4, I aim to analyse the possible data process landscape, advise my stakeholders regarding the various opportunities and the possible trade-offs, make a design of the architecture possibilities, realize proof of concept for the skin-cancer AI pipeline, and manage/control its evolution with measurable acceptance criteria. At Level 3, I will try to strengthen my knowledge regarding technical requirements analysis, software design, and delivery practices for AI-enabled systems, with the clear goal of producing a reliable, testable prototype, and potentially an improved ai model with some of the known biases covered and accounted for.

Proof of learning outcomes

I will evidence each outcome with concrete artefacts linked to my research project and relevant coursework.

Organisational Processes	
Learning outcome	How will I show proficiency (semester 1)
Level 4 - Analysis	I will aim to frame the complex dermatology-AI context; run literature review + EDA to surface problems/opportunities that generalize (bias, labeling cost, etc.)
Level 4 - Advise	I will aim to create practice-oriented advice for the stakeholders by multi-criteria decision matrix, plus a competitive academic literature scan to justify choices.
Level 4 - Design	I will aim to produce a generalizable architecture design focusing on well documented and widely regarded best practices.
Level 4 - Realize	I will try build an end-to-end pipeline POC, covering the steps of synthetic data generation, semi-automated labeling whenever applicable, and documented planned experiments with various datasets (fully synthetic data, and different hybrid spreads of real and synthetic data)
Level 4 - Manage & Control	I will aim to implement a standardized way of working for the pipeline employing locked splits, predefined metrics (e.g. ROC-AUC), benchmark protocol, cost/annotation tracking, and cadence to milestones.

Software	
Level 3 - Analysis	I will aim to define module-level requirements and acceptance thresholds; specify experiment/evaluation protocol and maintain an experiment backlog.
Level 3 - Advise	I will try to recommend concrete module/tech choices with cost/performance rationale and a decision log.
Level 3 - Design	I will try to design modules for data loaders with origin flags, ratio scheduler, prompt templates, and quality-filter logic, versioned configs for runs.
Level 3 - Realize	I will try to implement modules and tests, a reproducible repo and execute baselines and candidates with fixed seeds when applicable, reported confusion matrices and domain-gap analyses when applicable.
Level 3 - Manage & Control	I will try to establish CI, code quality checks if applicable, experiment logs, semantic versioning.

Professional Skills	
Professional Standard	I will operate ethically and responsibly by adhering to GDPR compliance by design, fairness/bias reporting and by making sustainable future-oriented choices.
Personal Leadership	I shall seek constant feedback and development input from experts and supervisors, and document reflections, learning goals, and learning process aligned to my target final development goal.
Academic Standard	I will demonstrate this mastery by making use of an extensive literature review, employment of applied statistics for validation.

Learning goals for the second semester

Semester 2 will be an internship, where I would like to focus on data analytics and digital transformation. My aim is to translate real business questions into usable data products, strengthen stakeholder communication, and ship production-ready dashboards and analyses. Typical responsibilities for the roles I'm targeting include supporting dashboard creation, driving digital-change adoption, investigating new tools/trends, contributing to digital-strategy meetings, and collaborating with digital ambassadors, a strong fit with my skills in SQL/Power BI and my growth goals.

Proof of learning outcomes

Organisational Processes	
Learning outcome	How will I show proficiency (semester 2)
Level 4 - Analysis	In general, I would like to show my proficiency by doing some form of a company discovery, such as stakeholder map, as-is process/analytics landscape, KPI baseline, etc.
Level 4 - Advise	I would like to show my proficiency by giving executive brief with options and TCO/ROI recommendation on visualisation standards and adoption plan, inputs to digital-strategy sessions with change impacts.
Level 4 - Design	I would like to show my proficiency by showing target analytics operating model, semantic data model, and rollout roadmap aligned to business priorities.
Level 4 - Realize	I would like to deliver something like 2–3 production-grade dashboards backed by well-tested SQL models, run trainings, measure adoption (usage, time-saved) and before/after KPIs.
Level 4 - Manage & Control	I would like to set up a standardised way-of-working, within a company, something amongst the lines of setting up backlog to release flow, naming conventions, data dictionary, governance cadence, and incident/change workflows embedded with “digital ambassadors.”

Software	
Learning outcome	How will I show proficiency (semester 2)
Level 3 - Analysis	In general, I want to find elicited requirements and SLIs, profile datasets, write performant SQL; define acceptance criteria for each dashboard/use case.
Level 3 - Advise	I want to recommend tool/stack patterns (e.g., data gateway, workspace model) with cost-performance trade-offs, maintain a clear decision log.
Level 3 - Design	I would want to try and do something like a Star/denormalised schema for analytics, metric definitions, and wireframes and possibly define row-level security and refresh schedules.
Level 3 - Realize	I want to build SQL views/models and reports, implement tests/validations, publish to workspaces, set alerts and basic monitoring.
Level 3 - Manage & Control	I want to try and set up versioning for data models/reports, release notes, quality gates, and usage dashboards to inform continuous improvement.

Professional Skills	
Learning outcome	How will I show proficiency (semester 2)
Professional Standard	Operate with GDPR/privacy-by-design, ethical data use, accessibility in visuals, and sustainability considerations; capture required signoffs.
Personal Leadership	Internship OKRs, feedback cycles, facilitation of stakeholder sessions/training, and documented reflections on career direction in data & analytics.
Academic Standard	Reflective report linking practice to literature on BI adoption/visual analytics; transparent methods and reproducible artefacts for assessment.

Self-assessment

This chapter will be focusing my first self-assessment/s against the programme's learning outcomes. Using the development-oriented scale, Undefined, Orienting, Beginning, Proficient, Advanced, which is outlined in appendix A, I will rate my current performance and note initial evidence and gaps. The ratings reflect artefacts completed so far (e.g., literature review notes, EDA, early PoC concepts), feedback from my coaches and stakeholders, and early metrics where applicable.

Self-assessment at study week 5

By the checkpoint in study week 5, I have made a start and explored concrete ways to demonstrate each learning outcome. I scoped out the project and began with initial opening steps such as a literature review and opening talks with stakeholders. Additionally, I ran 3 general experiments to examine the possibilities of having data pipelines with basic open-source available models, with decent initial results.

Learning outcome	Latest assessment by coach	Self-assessment
Level 4 - Analysis		Orienting
Level 4 - Advise		Orienting
Level 4 - Design		Orienting
Level 4 - Realize		Orienting
Level 4 - Manage & Control		Orienting
Level 3 - Analysis		Orienting
Level 3 - Advise		Orienting
Level 3 - Design		Orienting
Level 3 - Realize		Orienting
Level 3 - Manage & Control		Orienting
Professional Standard		Orienting
Personal Leadership		Orienting
Academic Standard		Orienting

Planning

The planning section provides a roadmap for the completion of my research project for this semester, within a structured 17-week timeframe. The planning gives an overview of the applied research activities, which seek to achieve the delivery of a practical research outcome that advances the application of dermatological AI models supported by synthetic data, and the academic knowledge within that domain.

Project Timeline

Phase	Description
Phase 1: Literature Review and Benchmark Analysis [Weeks 1-3]	• Comprehensive analysis of ISIC challenge evolution (2016-2018) and task progression • Dataset characterization: HAM10000 7-category analysis, BCN20000 "in-the-wild" challenges, Fitzpatrick 17k bias patterns • Review of Raumanns et al.'s multi-task learning and crowdsourced annotation methodologies • Analysis of patient-centric approaches from Rotemberg et al.'s clinical context integration • Semi-supervised learning and synthetic data generation literature review
Phase 2: Solution Design and Prototype Development [Weeks 4-8]	• Design synthetic data generation pipeline (API-based approach vs. fine-tuning evaluation) • Develop semi-automated annotation workflow using baseline/weakly-trained models • Create end-to-end pipeline (data generation to model evaluation) • Implement quality filtering system for "hallucination" detection in synthetic data • Prompt engineering optimization for domain-specific generation

Phase 3: Experimental Implementation and Validation [Weeks 9-12]	• Conduct systematic experiments with multiple dataset compositions (real-only, synthetic-only, hybrid ratios) • Implement comprehensive validation using COCO metrics, confusion matrices, and domain gap analysis • Measure annotation time reduction and cost-efficiency metrics • Validate model collapse prevention through hybrid training strategies • GDPR compliance assessment for synthetic data generation
Phase 4: Documentation and Reporting [Weeks 13-17]	• Research paper writing • Technical documentation for handover • Research poster preparation • Code quality assurance and documentation

Key Milestones and Deadlines

Critical Milestone	Completion time
Literature review completion	Week 3
Prototype proof-of-concept completion	Week 8
Validation framework implemented	Week 10
Experimental results and analysis	Week 12
First draft of final paper	Week 14
Technical documentation completion	Week 15
Final paper completion	Week 16
Project poster completion and presentation	Week 17
Project handover	Week 17

Appendix A: Development-oriented feedback scale

Formative indication	Explanation
Advanced	You have shown several times that you have been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than sufficient level, if you continue your development in this way.
Proficient	You have shown several times that you have created a basis to demonstrate the learning outcome. You will demonstrate the learning outcome at a sufficient level, if you continue your development in this way.
Beginning	You have taken the first steps and carried them out which contribute to demonstrating the learning outcome.
Orienting	You have made a start and explored the possibilities to demonstrate the learning outcome.
Undefined	You have not yet undertaken activities to demonstrate the learning outcome.

Appendix B: Overview of learning outcomes

Learning outcome	Explanation
Analysis	A complex multidisciplinary context in professional practice to identify problems and opportunities for the use of IT that generalize to other contexts
Advise	Advise on the future use of IT in professional practice based on solid practice-oriented research.
Design	Design a generalizable IT solution or framework based on solid practice-oriented research to address a problem or opportunity in professional practice
Realize	Realize: build a generalizable IT solution or framework that addresses a problem or opportunity in professional practice
Manage&control	Manage & Control: design and implement a standardized way-of-working in professional practice including the necessary IT support
Professional Standard	You take responsibility when solving an ICT issue. You define and carry out your research using relevant selective methodologies and provide advise to your stakeholders even in complex and uncertain contexts. You substantiate and validate future oriented choices by use of ethical, intercultural, and sustainable arguments.
Personal leadership	You are enterprising around IT assignments and personal development, pay attention to your own learning ability and keep in mind what kind of IT professional and/or what type of positions you aspire to be/fulfill.
Academic Standard	Your final level of your Masters programme is distinguished by mastery, investigative ability, interprofessionalism, and impact.

For example descriptions of learning outcomes per domain/architecture layer, please refer to https://www.hbo-i.nl/wp-content/uploads/2024/05/24040_HBOi_Domeinbeschrijving_ENG.pdf